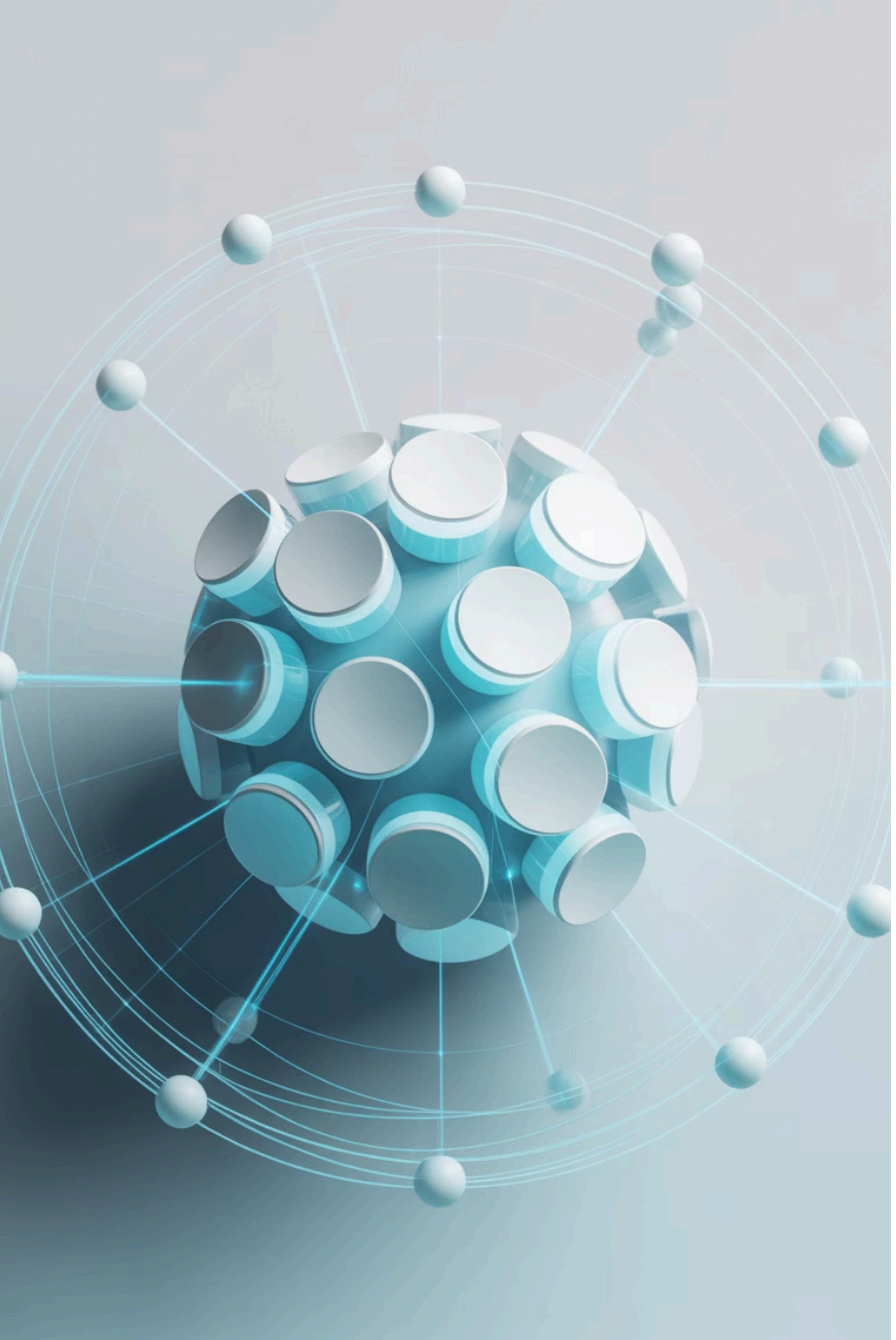
Multi-Agent Systems (MAS)

Roles, Deliveries, Context & Collaboration

Welcome to the foundation of collaborative AI systems. In this section, we'll explore how multiple specialized agents work together to solve complex problems that single agents cannot handle alone.



Core Principles of Multi-Agent Systems

One Agent = One Role

Each agent specializes in a specific function, just like team members in a successful organization. This specialization ensures expertise and clarity.

Define Role, Delivery, Context

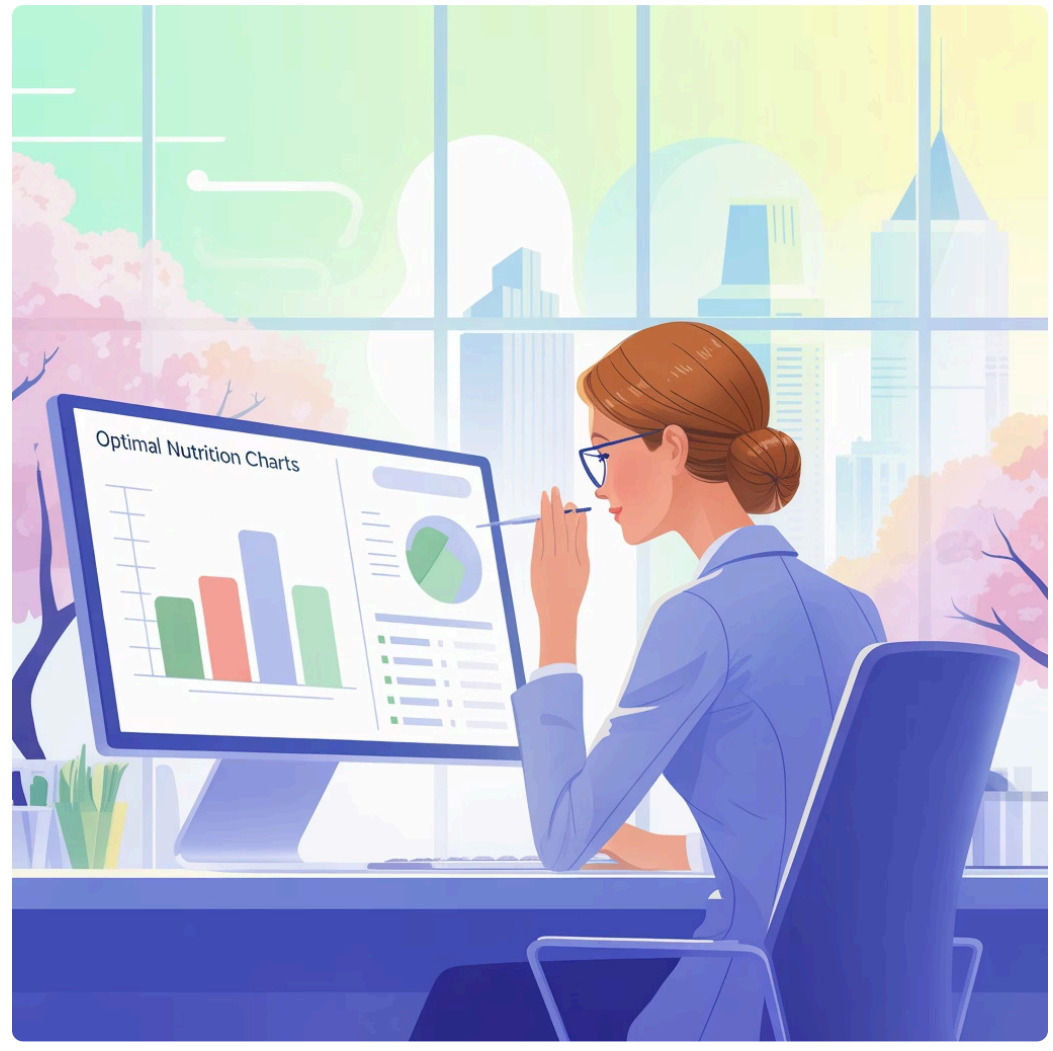
Every agent needs a clear job description: what they do (role), what they produce (delivery), and their working environment (context).

Collaborate via Hand-offs

Agents work together by passing outputs as inputs, creating a seamless workflow that builds upon each agent's expertise.

These principles mirror successful human teams: clear roles, defined deliverables, and structured communication create powerful collaborative systems.

Real-World Example: Chef + Nutritionist Demo



The Chef Agent

Role: Creative recipe developer

Input: Single ingredient (e.g., "chicken")

Delivery: 5 diverse, detailed recipes with cooking instructions, prep time, and serving suggestions

Context: Culinary expertise with focus on variety and flavor

The Nutritionist Agent

Role: Health-focused food analyst

Input: Chef's 5 recipes

Delivery: Best recipe recommendation for diabetics with detailed nutritional reasoning

Context: Medical nutrition knowledge, diabetic dietary restrictions

The Magic of Collaboration: The Chef creates without nutritional constraints, while the Nutritionist applies specialized health knowledge. Neither could produce the final result alone—the Chef lacks medical expertise, and the Nutritionist lacks culinary creativity.

Customer Service Multi-Agent System

L1 Support Agent

Role: Research and draft comprehensive responses

Tools: Knowledge base search, documentation access

Delivery: Thorough, fact-based draft reply with supporting information

L2 Quality Agent

Role: Review, refine, and polish responses

Tools: None needed—focuses purely on communication quality

Delivery: Professional, empathetic, ready-to-send customer response

Key Insight: Keep tool count minimal to avoid coordination overhead. L1 handles research-heavy tasks with tools, while L2 focuses on human communication without tool distractions.

When NOT to Add More Agents

Coordination Costs Add Up

Each additional agent increases system complexity exponentially. More agents mean more potential failure points, longer processing times, and higher computational costs.

Only Add Truly Distinct Roles

Ask yourself: "Does this agent bring a fundamentally different skill set?" If two agents could reasonably do each other's jobs, consolidate them into one more capable agent.

Clear Prompts Trump More Agents

Before adding complexity, optimize what you have. Well-crafted prompts and detailed backstories often solve problems better than adding more agents to the system.

> "The best multi-agent system is the simplest one that still accomplishes your goals. Every agent should earn its place through unique, irreplaceable value."